

— PROPOSAL DEFENSE

Interactive AI Literacy Simulation for Junior High School Students

Frontend interaction design, Top-3 UI, and game-based Human-in-the-Loop experience.

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PROGRAM

Applied Bachelor of Multimedia Engineering
Technology

INSTITUTION

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Why This Project Is Needed

AI literacy should not only teach students how to use AI, but also how to evaluate AI output.

PROBLEM	IMPACT ON STUDENTS	PROPOSED DIRECTION
AI output is often treated as a final answer.	Students may accept AI without evaluation.	Show AI prediction as uncertain output.
AI concepts are abstract for younger learners.	Students may not understand confidence and error.	Use drawing-based interactive simulation.
Learning media often lacks decision consequence.	Students only observe, not decide.	Add a Human-in-the-Loop decision moment.

AI

is introduced as a probabilistic system.

HITL

keeps students as active evaluators.

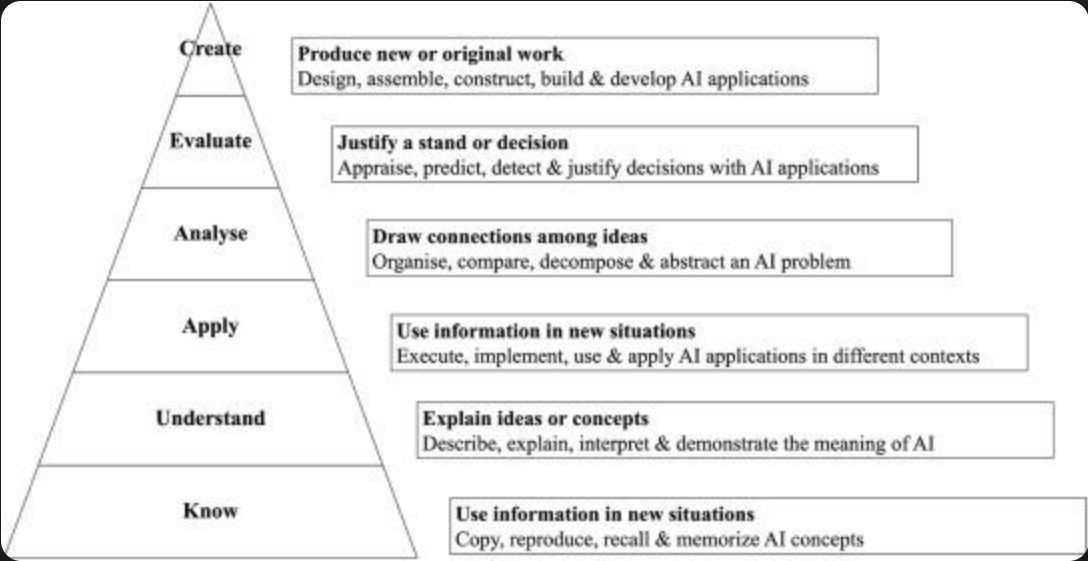
Game

turns decisions into visible consequences.

Log

records interaction patterns for review.

Inspired by exploratory AI literacy approaches for younger learners, but implemented through visual classification and gameplay decisions.



AI literacy dimensions used as the educational foundation.

Sketch, Predict, Decide, and See the Consequence

A web-based interactive simulation where students draw objects, inspect AI predictions, and decide the final object behavior.

Student Drawing

AI Prediction

Top-3 UI

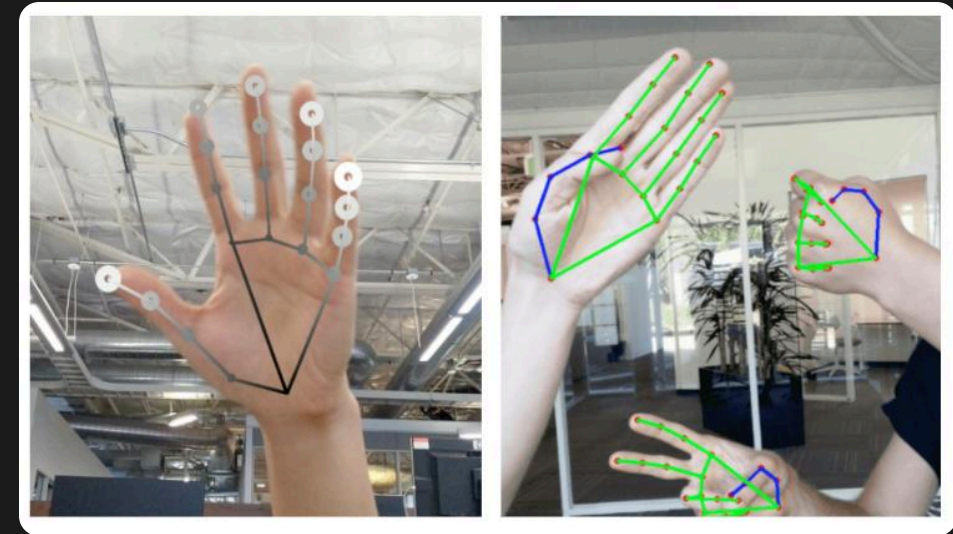
Human Decision

2D Gameplay
Consequence

Interaction Log

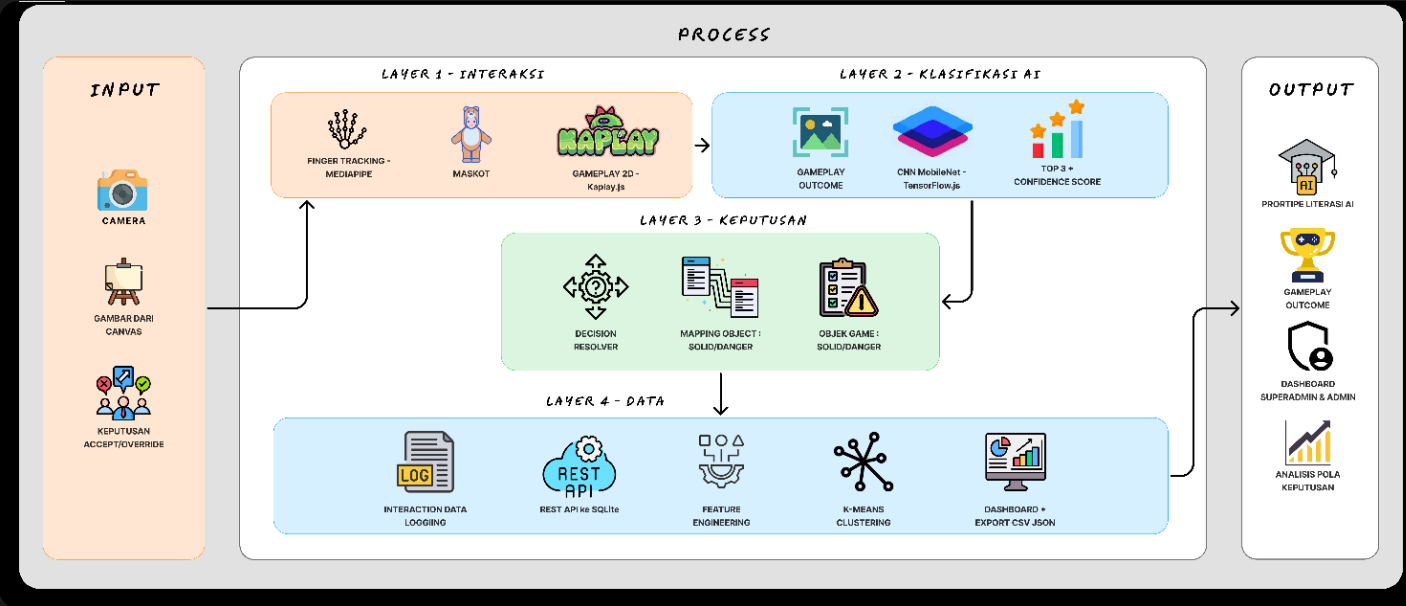
MAIN POINTS

- Students draw objects through a canvas-based interaction.
- The system displays AI prediction through a Top-3 UI.
- Students choose to accept, correct, override, or redraw.
- The decision affects object behavior inside the 2D game.



One Integrated Global System

The system connects finger tracking input, AI prediction, Human-in-the-Loop validation, gameplay behavior, and interaction data.



Input

Camera stream, finger tracking, canvas drawing, and student decision.

Process

Interaction layer, AI classification, decision resolver, and log handling.

Output

Prediction UI, gameplay outcome, learning feedback, and decision pattern data.

Author Scope Within the Team Project

This proposal focuses on frontend interaction, UI/UX, Top-3 decision flow, gameplay, and the student-facing learning experience.

AREA	AUTHOR SCOPE	PARTNER RELATION
User Interface	Main screens, game HUD, Top-3 UI, and feedback display.	Receives prediction data and log structure.
Interaction	Finger tracking, canvas drawing, navigation, and decision input.	Uses interaction output as recorded data.
Gameplay	2D level flow, object behavior, fail state, and retry condition.	Uses the AI prediction result as gameplay input.
Feedback	Mascot response, level result, and decision summary.	Supports data analysis after interaction.
Documentation	User flow, interaction matrix, wireframe, and level interaction design.	Integrated into the global system proposal.

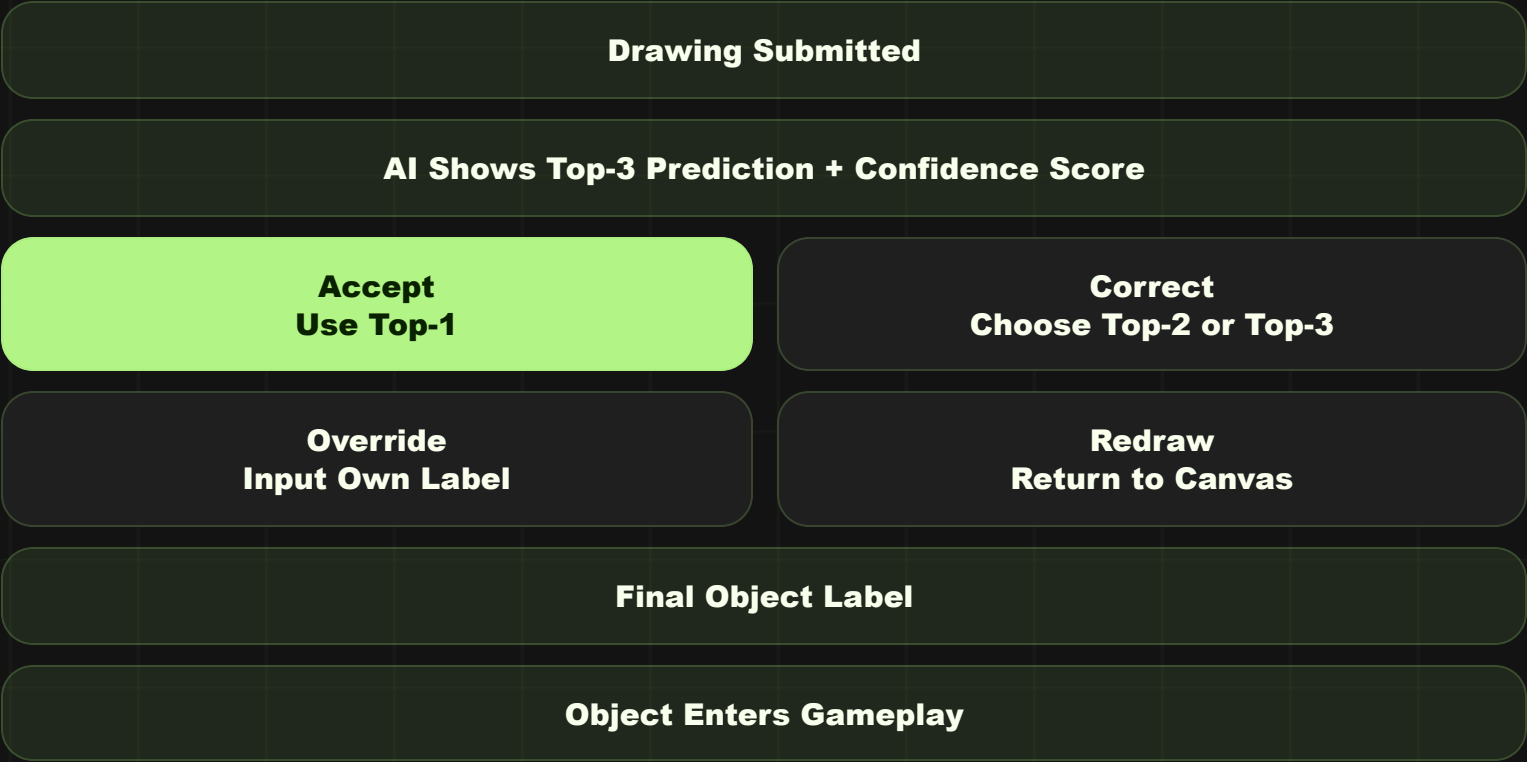
Mapping Input, Response, and Simulation Context

The interaction matrix explains how each student action creates a clear system response.

ID	INTERACTION TYPE	USER INPUT	SYSTEM RESPONSE	USED IN
I-01	Presence	Wave hand	System starts and companion greets.	Onboarding
I-02	Navigation	Click, tap, or gesture	Start, continue, or next screen.	Menu and tutorial
I-03	Drawing	Canvas drawing	Drawing is captured.	Core loop
I-04	AI Request	Submit drawing	Top-3 prediction appears.	Top-3 UI
I-05	AI Decision	Accept, correct, override, or redraw	Final label is selected.	HITL moment
I-06	Movement	Keyboard, touch, or gesture	Main character moves.	Main gameplay
I-07	Feedback	User action result	Feedback and consequence appear.	Result state
I-08	Recovery	Retry or redraw	User revises action.	Fail state
I-09	Reflection	View summary	Decision summary appears.	End level

The Main Human-in-the-Loop Checkpoint

The student does not directly accept AI output. The Top-3 UI creates a decision point before the object enters gameplay.



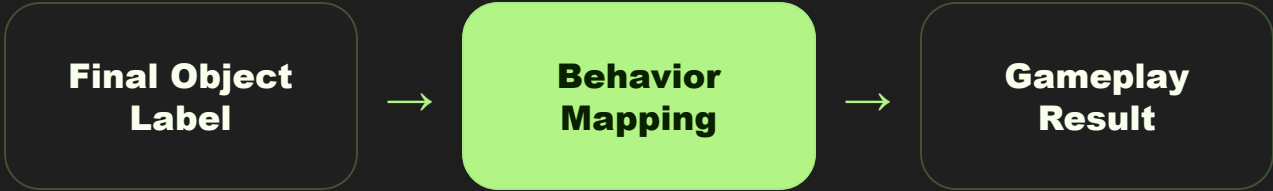
DECISION	LEARNING ROLE
Accept	Tests student trust toward AI output.
Correct	Tests comparison of alternative predictions.
Override	Tests human agency over AI suggestions.
Redraw	Tests recovery behavior when input is unclear.

Student Decisions Become Gameplay Behavior

The final label is translated into object behavior inside the simulation world.

FINAL LABEL RESULT	GAMEPLAY CATEGORY	BEHAVIOR IN SIMULATION	EXAMPLE
Safe support object	Solid	Can be used as a platform or helper object.	Ladder, bridge, stairs
Harmful object	Danger	Causes fail condition or blocks progress.	Fire, lightning, knife
Invalid or unclear object	Redraw or correction	Student must revise the drawing or label.	Unclear sketch

BEHAVIOR FLOW



- Solid objects help the student progress.
- Danger objects create visible consequences.
- Unclear objects guide the student back to correction.

Progressive AI Uncertainty Across Levels

Each level introduces a different intensity of AI uncertainty and student decision-making.

LEVEL	INTERACTION FOCUS	AI DISPLAY	STUDENT TASK	MAIN LEARNING MOMENT
Level 1	Guided recognition	Top-1 only	Draw and accept a simple object.	Understand that AI can classify sketches.
Level 2	Ambiguous prediction	Top-3 + confidence	Compare prediction alternatives.	Learn that AI output has uncertainty.
Level 3	Critical decision	Confident but wrong prediction	Override or redraw.	Learn that humans must validate AI.

1

Guided

Students learn the basic loop with low ambiguity and simple feedback.

2

Ambiguous

Students compare alternatives and interpret confidence values.

3

Critical

Students challenge AI output when the prediction does not match their intention.

Create, Decide, and See the Consequence

The experience is organized into three student-facing phases and supported by admin roles.

1

Create

Student enters the level, draws an object, and submits the sketch.

2

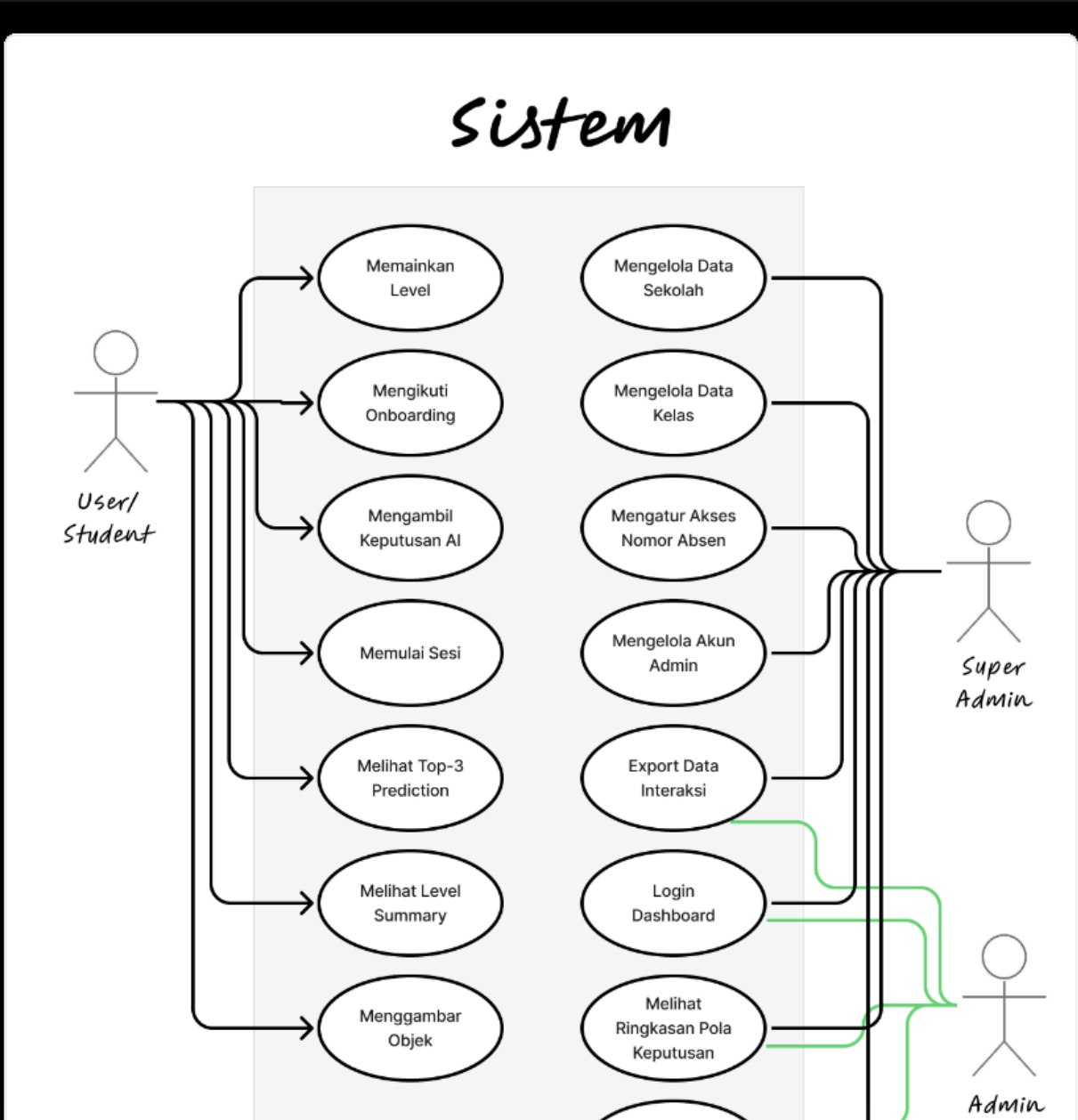
Decide

Top-3 UI appears and the student chooses Accept, Correct, Override, or Redraw.

3

Consequence

Object behavior is applied and the student receives feedback through gameplay.



Interaction First, Education Through Play

The mascot bridges the system hierarchy so students learn from the experience, not from a lecture.

Interaction

Natural entry point through camera, gesture, drawing canvas, and simple UI.

Gameplay & Storyline

A simple puzzle-platformer context turns drawing and prediction into a playable goal.

Education

AI literacy appears through reflection on prediction, confidence, correction, and consequence.



Mascot as Bridge

A visual companion that softens instruction, signals uncertainty, and connects game feedback to AI literacy.

DESIGN PRINCIPLE

Teach without making students feel lectured.

- Interaction lowers friction so students can start playing before facing theory.
- Gameplay gives students a reason to care about their AI decision.
- The mascot turns prediction moments into short cues, warnings, and reflections.
- Education is placed at the end of the loop, after students experience the consequence.

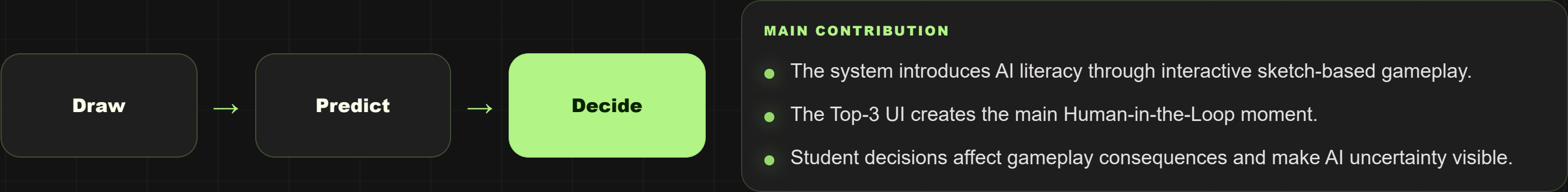
DESIGN INSPIRATION

MIT Little Language Models / CoCo shows how AI concepts can be introduced to younger learners through playful exploration. This project adapts the interaction-first idea into a visual classification context: students draw, inspect Top-3 predictions, decide, and see gameplay consequences.

Research Schedule and Main Contribution

The project plan follows the development phases while keeping the main contribution focused on interactive AI literacy.

ACTIVITY	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
Concept	■	■										
Design		■	■	■								
Material Collecting				■	■	■						
Assembly					■	■	■	■				
User Testing								■	■	■		
Distribution										■	■	■



— CLOSING

Thank You

I am open to questions, feedback, and suggestions.